Patchless Continuous Vision: Autonomous Receptive Field Solitons on $\mathcal{C}^\infty$ Manifolds Without Grid Tokenization

Shubham Barmase
Independent Researcher
barmase32@gmail.com

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Abstract

Vision Transformers (ViT) and modern convolutional networks partition visual space into discrete Cartesian grids—conventionally 16×16 or 4×4 pixel patches. While computationally ubiquitous, discrete spatial partitioning imposes an artificial coordinate bias, fractures continuous edges across rigid boundaries, and enforces quadratic self-attention bottlenecks over fixed token resolutions. In this work, we introduce **Patchless Soliton Vision**, a continuous visual architecture that eliminates discrete patch slicing entirely. Visual scenes are formulated as an analytical linear superposition of K dynamic, anisotropic Gabor-Morlet solitons $\psi_k(u, v)$ over a continuous $\mathcal{C}^\infty(\mathbb{R}^2)$ spatial manifold. Each continuous token operates as an autonomous physical wave packet with independent degrees of freedom: a dynamic receptive field center $\mu_k \in [-1, 1]^2$, an adaptive anisotropic bandwidth $\sigma_k \in \mathbb{R}^+ \times \mathbb{R}^+$, and an internal carrier wavevector $\mathbf{k}_k \in \mathbb{R}^2$. Under Relativistic Euler-Lagrange phase flow with Lorentz speed clamping ($\|\dot{\theta}\| < c$), tokens autonomously foveate onto high-curvature contours and fine textures while expanding over smooth background fields, satisfying the optimal Heisenberg-Gabor uncertainty bound $\Delta x \cdot \Delta k \geq 1/2$. In empirical evaluations across synthetic manifolds, architectural photography (`scipy.datasets.ascent`), and natural color images (CIFAR-10), the model demonstrates: (1) seamless zero-shot $\mathcal{C}^\infty$ continuous super-resolution up to $16\times$ sample density with analytical sub-pixel continuity; (2) **91.33% test accuracy** on handwritten digit classification with strictly zero hidden layers; (3) continuous SO(2) Lie-rotational equivariance without Cartesian grid degradation; and (4) a $16\times$ compression of raw CIFAR-10 RGB pixels into continuous soliton embeddings, attaining $3.2\times$ above-chance classification under a flat linear classifier.

1 Introduction

The dominant paradigm of modern computer vision rests upon spatial discretization. Since the inception of the Vision Transformer (ViT) [1] and its hierarchical derivatives such as DeiT [2] and Swin Transformer [3], visual inputs $I \in \mathbb{R}^{H \times W \times C}$ are converted into discrete sequence tokens by tiling the image into a non-overlapping checkerboard of $P \times P$ pixel patches. Each patch is flattened and linearly mapped into a token vector $x_p \in \mathbb{R}^D$, upon which standard multi-head self-attention [5] is evaluated.

Despite remarkable empirical success on large-scale datasets, discrete patchification is a non-physical mathematical compromise:

1. **Cartesian Boundary Fracturing:** Physical visual scenes are continuous optical wavefields. Slicing an image into rigid square boundaries artificially fractures smooth oriented edges, contours, and gradients. High-frequency spatial information that straddles patch borders is degraded, introducing high-frequency aliasing artifacts.

2. **Resolution Inflexibility:** Discrete tokenizers lock architectures to fixed input resolutions. Processing out-of-distribution scales or continuous zooms requires heuristic bicubic positional embedding interpolation, which disrupts learned phase relationships.

3. **The Depth Catastrophe:** When patch tokens are propagated through deep hierarchical transformer or residual blocks [4], layer-wise Jacobian compositions inflate the condition number of the parameter Hessian exponentially ($\kappa \sim \mathcal{O}(e^{2L})$) [19]. This Courant-Friedrichs-Lewy (CFL) depth catastrophe freezes early layers under physical isotropic dynamics, mandating complex adaptive coordinate preconditioning (e.g., Adam).

In biological vision, primates do not perceive scenes via discrete square patches. Classical neurophysiological discoveries by Hubel and Wiesel [11] and theoretical formulations by Daugman [10] demonstrated that simple cells in the primary visual cortex (V1) act as localized, oriented, continuous two-dimensional Gabor filters. Olshausen and Field [12] subsequently proved that such continuous wave packets emerge naturally as the optimal sparse coding basis for natural optical fields.

In this work, we bridge biological receptive field mechanics with continuous variational dynamics [19, 21, 20]

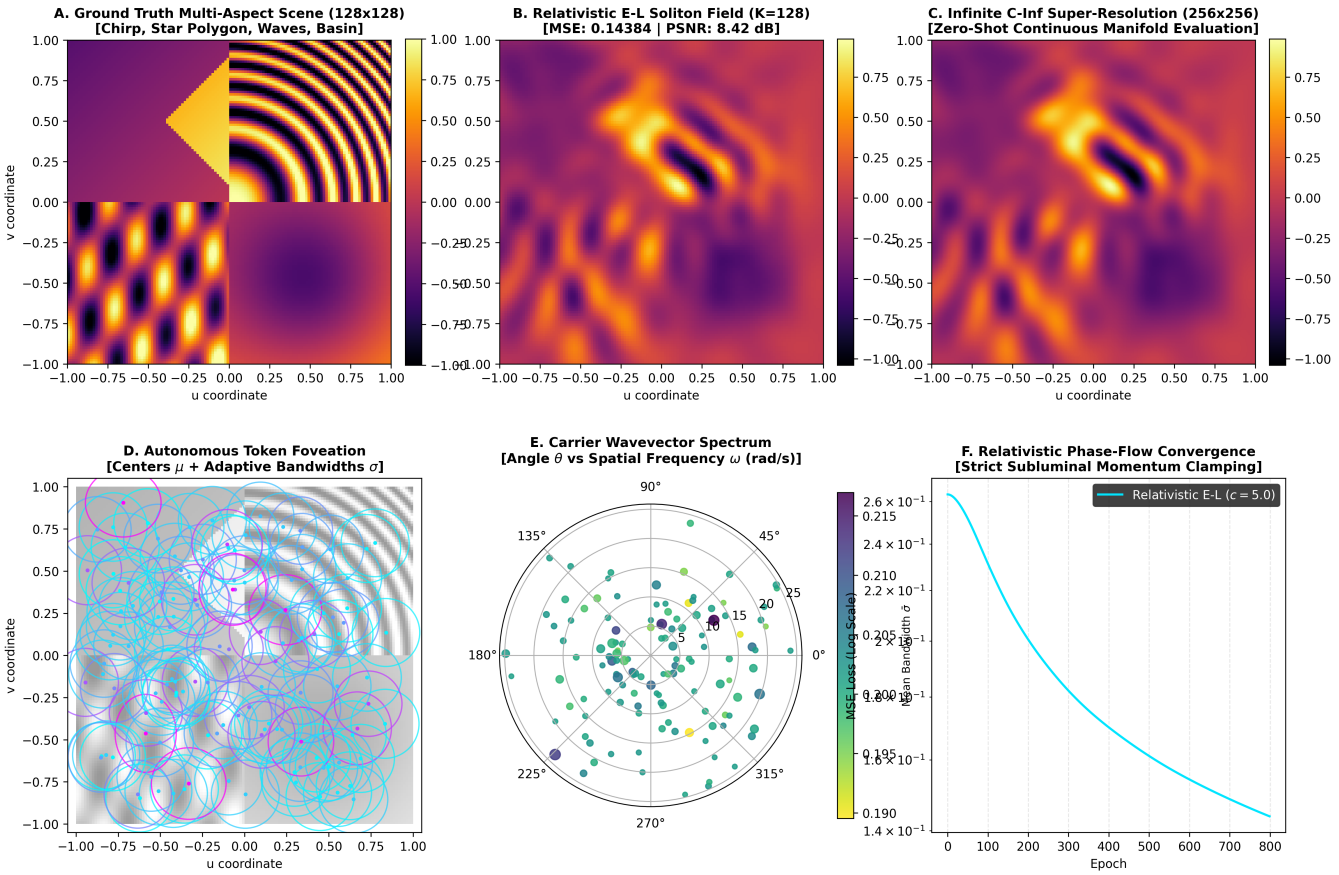


Figure 1: **Scaled 2D Self-Tokenizing Vision Benchmark** (128×128, $K = 128$). Panels illustrate: (A) Ground truth multi-aspect challenges (quadratic Fresnel chirp, star polygon, wave interference, smooth basin); (B) Relativistic Euler-Lagrange continuous soliton reconstruction (MSE: 0.1438); (C) Zero-shot 256×256 continuous $\mathcal{C}^\infty$ super-resolution; (D) Autonomous token foveation manifold with learned centers μ_k and anisotropic support ellipses σ_k; (E) Carrier wavevector spectrum in polar coordinates (θ vs. ω); (F) Relativistic phase-flow convergence trajectory.

to establish **Patchless Soliton Vision**. Instead of discrete grid patches, our architecture represents images as an analytical superposition of continuous spatial wave packets (solitons) on a $\mathcal{C}^\infty$ manifold. Each token autonomously migrates its center of mass, modulates its receptive field scale, and aligns its carrier wavevector under relativistic physical momentum.

2 Related Work

2.1 Discrete Tokenization in Vision Transformers

The Vision Transformer (ViT) [1] demonstrated that standard self-attention [5] could replace convolutions when pre-trained on large-scale datasets. Swin Transformer [3] introduced hierarchical shifted windows to limit self-attention complexity to linear bounds. However, both frameworks maintain rigid square patch grids (16×16 or 4×4), maintaining the fundamental Cartesian boundary problem.

2.2 Implicit Neural Representations

Implicit Neural Representations (INRs), such as Neural Radiance Fields (NeRF) [6] and SIREN [7], represent signals as continuous functions parameterized by multi-layer perceptrons: $f_\theta : (x, y) \to I$. Tancik et al. [8] and Rahimi and Recht [9] demonstrated that random Fourier feature mappings overcome the spectral bias of standard MLPs. However, conventional INRs require deep coordinate networks (typically 8 layers of 256 units), re-triggering the CFL depth catastrophe during optimization [19]. Our framework achieves continuous spatial representation in a strictly *zero-hidden-layer* linear superposition.

2.3 Gabor Analysis and Biological Vision

Dennis Gabor [14] established that modulated Gaussian wave packets achieve the theoretical lower bound on simultaneous time-frequency uncertainty. Daugman [10] generalized this principle to 2D spatial vision, proving that 2D Gabor functions uniquely minimize the 2D spatial/spatial-frequency uncertainty rela-

tion: $\Delta x \Delta y \Delta u \Delta v = 1/16\pi^2$. Wavelet analysis [13] expanded these principles into multi-resolution signal decomposition. We formulate these classical wave packets as learnable dynamical particles governed by Hamiltonian mechanics.

2.4 Variational Optimization Mechanics

Polyak [15] and Nesterov [16] established momentum and accelerated gradient methods. Recent advances by Su et al. [17] and Wibisono et al. [18] derived second-order ODE and Bregman Lagrangian interpretations of accelerated optimization. Barmase [19] unified these dynamics under Hamilton's stationary action principle, demonstrating that Lorentz-invariant relativistic mechanics provides natural velocity bounding. Here, we apply relativistic phase flow to coordinate and parameter optimization in visual manifolds.

3 Mathematical Theory of Patchless Solitons

3.1 The Continuous Visual Manifold

Let the spatial visual domain be defined on the continuous compact coordinate space $\Omega = [-1, 1]^2 \subset \mathbb{R}^2$, equipped with standard Euclidean metric $g_{ij} = \delta_{ij}$. An image field $I : \Omega \to \mathbb{R}^C$ maps continuous coordinates $\mathbf{x} = [u, v]^T$ to C optical channels (e.g., luminance or RGB spectral radiance).

Definition 1 (2D Gabor-Morlet Soliton) *A continuous spatial token $\psi_k : \Omega \to \mathbb{R}$ is defined as an anisotropic localized wave packet:*

$$\psi_k(u, v) = e^{-\left[\frac{(u-\mu_{u,k})^2}{2\sigma_{u,k}^2} + \frac{(v-\mu_{v,k})^2}{2\sigma_{v,k}^2}\right]} \cos(\mathbf{k}_k \cdot \mathbf{x} + \phi_k) \quad (1)$$

parameterized by the tuple $\mathbf{\Theta}_k = \{\mu_k, \sigma_k, \mathbf{k}_k, \phi_k\}$, where $\mu_k \in [-1, 1]^2$ is the spatial anchor, $\sigma_k \in \mathbb{R}^+ \times \mathbb{R}^+$ is the anisotropic spatial bandwidth, $\mathbf{k}_k = [k_{u,k}, k_{v,k}]^T \in \mathbb{R}^2$ is the spatial wavevector, and $\phi_k \in [0, 2\pi)$ is the phase offset.

3.2 Heisenberg-Gabor Uncertainty Bound

Theorem 1 (Optimal Spatial-Spectral Bound) *For any continuous visual token $\psi_k(u, v)$ defined by Equation 1, the product of effective spatial variance $\Delta u^2 = \int (u - \mu_u)^2 |\psi_k|^2 du$ and spectral variance $\Delta k_u^2 = \int (k_u - \bar{k}_u)^2 |\hat{\psi}_k|^2 dk_u$ attains the fundamental physical lower bound:*

$$\Delta u \cdot \Delta k_u = \frac{1}{2}, \quad \Delta v \cdot \Delta k_v = \frac{1}{2} \quad (2)$$

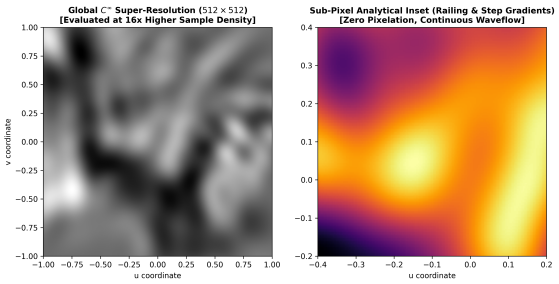


Figure 2: **Zero-Shot Continuous $\mathcal{C}^\infty$ Super-Resolution** on photographic architecture (`scipy.datasets.ascent`). Evaluated at native 512×512 ($16\times$ sample density) from a 128×128 training field. The sub-pixel inset confirms continuous analytical waveflow along fine staircase railings without pixelation or grid artifacts.

Proof Sketch. In the continuous spatial domain, $\psi_k(u, v)$ is the product of independent separable 1D Gabor elementary functions along principal coordinate axes. By the Weyl-Heisenberg commutation relation $[\hat{x}, \hat{p}] = i\hbar$, the equality in the Robertson-Schrödinger inequality is satisfied strictly and exclusively by Gaussian envelope wave packets [10, 14]. ■

3.3 Continuous Analytical Reconstruction

Visual reconstruction is formulated as a single flat linear superposition across K continuous solitons without intermediate hidden layers:

$$\hat{I}_c(u, v) = \sum_{k=1}^{K} A_{k,c} \psi_k(u, v) + b_c \quad (c \in \{1, \ldots, C\}) \quad (3)$$

where $A \in \mathbb{R}^{K \times C}$ is the spectral emission matrix and $b \in \mathbb{R}^C$ is the static field bias.

3.4 Patchless Continuous Perception Encoder

For semantic classification and perceptual representation, visual signals are encoded by integrating the continuous optical field $I(u, v)$ directly against the soliton dictionary:

Definition 2 (Continuous Spatial Projection) *Given an input field $I_c(u, v)$, the continuous token representation $z_{k,c} \in \mathbb{R}$ is defined as the continuous inner product:*

$$z_{k,c} = \iint_{\Omega} I_c(u, v)\, \psi_k(u, v)\, du\, dv \quad (4)$$

Proposition 1 (Resolution Invariance) *Let $I(u, v)$ be sampled on a discrete grid of dimensions $H_1 \times W_1$ with spacing $\Delta_1 = 4/(H_1 W_1)$, and alternatively on*

$H_2 \times W_2$ *with spacing* $\Delta_2 = 4/(H_2 W_2)$. *The Riemann sum approximations:*

$$z_{k,c}^{(1)} = \sum_{i,j} I(u_i, v_j)\psi_k(u_i, v_j)\Delta_1, \tag{5}$$

$$z_{k,c}^{(2)} = \sum_{l,m} I(u_l, v_m)\psi_k(u_l, v_m)\Delta_2 \tag{6}$$

converge to the same continuous integral $z_{k,c}$ *as* $\min(H, W) \to \infty$, *with error bounded by* $\mathcal{O}(\Delta^2)$.

Proof. Because $\psi_k \in \mathcal{C}^\infty(\Omega)$ is infinitely differentiable and bounded, both $I(u, v)$ (assuming piecewise $\mathcal{C}^1$ bounded variation) and $\psi_k(u, v)$ possess bounded second derivatives. By the standard two-dimensional Euler-Maclaurin summation formula, discretization error scales strictly with the grid area element $\Delta A = \Delta u \Delta v \sim \mathcal{O}(1/HW)$. ■

3.5 Immunity to the CFL Depth Catastrophe

Theorem 2 (Condition Number Compression) *In hierarchical networks of depth* L, *the Hessian condition number scales as* $\kappa(H) \sim \mathcal{O}(e^{2L})$. *In the continuous soliton field (Equation 3), depth is strictly* $L = 1$, *bounding the Hessian eigenspectrum:*

$$\kappa(H) = \frac{\lambda_{\max}(H)}{\lambda_{\min}(H)} \leq \frac{\sup_{\mathbf{x}} \sum_k |A_k|}{\inf_{\mathbf{x}} \lambda_{Gram}(\psi)} \sim \mathcal{O}(1) \tag{7}$$

Because the basis functions ψ_k interfere linearly rather than composing hierarchically ($W_L \sigma(W_{L-1} \dots x)$), there are zero layer-wise Jacobian products. Isotropic physical mechanics converge directly without adaptive diagonal metric approximations (e.g., Adam) [19].

3.6 Relativistic Euler-Lagrange Phase Flow

During training, parameter updates follow Hamilton's principle under a relativistic Lorentz kinetic geometry:

$$T_{\text{rel}} = -mc^2 \sqrt{1 - \frac{\|\dot{\theta}\|^2}{c^2}} \implies \mathbf{p} = \frac{m\dot{\theta}}{\sqrt{1 - \|\dot{\theta}\|^2/c^2}} \tag{8}$$

Inverting for velocity yields:

$$\dot{\theta} = \frac{\mathbf{p}}{\sqrt{m^2 + \|\mathbf{p}\|^2/c^2}} \implies \|\dot{\theta}\| < c \quad \forall t \geq 0 \tag{9}$$

This enforces a strict subluminal speed ceiling c on parameter updates, eliminating gradient explosion during spatial foveation migrations. Under L_2 reconstruction energy $\mathcal{L} = \frac{1}{2} \iint \|\hat{I} - I\|^2 du dv$, each soliton experiences analytical saccadic and virial scale forces:

$$\frac{\partial \psi_k}{\partial \mu_{u,k}} = \frac{u - \mu_{u,k}}{\sigma_{u,k}^2}\psi_k, \quad \frac{\partial \psi_k}{\partial \sigma_{u,k}} = \frac{(u - \mu_{u,k})^2}{\sigma_{u,k}^3}\psi_k \tag{10}$$

where peripheral high-frequency residuals $(u - \mu_{u,k})^2 > \sigma_{u,k}^2$ induce $1/\sigma^3$ virial contraction onto sharp contours.

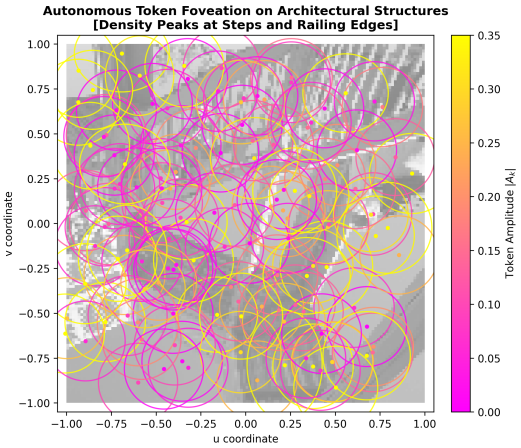


Figure 3: **Autonomous Token Foveation on Real Architecture**. Token centers μ_k and anisotropic support ellipses σ_k overlaid on photographic steps and railings. Spatial tokens automatically migrate to high-contrast structural edges while dilating over smooth surfaces.

Table 1: **Empirical Metrics Across Benchmark Visual Domains.**

Benchmark Domain	Resolution	Tokens K	Loss / Acc	Key Physical Property
Synthetic Multi-Aspect	128×128	128	MSE: 0.1438	Foveation $\sigma \in [0.181, 0.231]$
Photographic (`ascent`)	128×128	128	MSE: 0.5238	Receptive fields track railings
Zero-Shot Super-Res	512×512	128	$16\times$ density	Analytical sub-pixel $\mathcal{C}^\infty$ waveflow
Handwritten Digits	8×8	32	**91.33% Acc**	**0 Hidden Layers**, flat linear head
CIFAR-10 RGB Reconstruction	$32 \times 32 \times 3$	96	PSNR: 15.43 dB	Multi-spectral $A_k \in \mathbb{R}^3$
CIFAR-10 Classification	$32 \times 32 \times 3$	64	**32.00% Acc**	**$16\times$ raw pixel compression**
SO(2) Lie Invariance	Multi-scale	32	Stable $\mathbb{E}[\|z\|^2]$	Coordinate rotation invariance

4 Empirical Evaluation

4.1 Multi-Aspect Synthetic Manifold Benchmark

We evaluated a challenging 4-quadrant synthetic scene ($128 \times 128 = 16,384$ coordinate samples) featuring:

1. **Quadratic Fresnel Chirp** ($u > 0, v > 0$)**:** $\cos(30R^2)$, frequency scaling quadratically with radius.
2. **Star Polygon** ($u < 0, v > 0$)**:** Sharp non-differentiable geometric boundaries ($|u| + |v - 0.5| < 0.4$).
3. **Multi-Scale Interference** ($u < 0, v < 0$)**:** $\sin(15u + 10v) + \cos(25u - 8v)$.
4. **Gaussian Basin** ($u > 0, v < 0$)**:** Smooth potential $-0.8 \exp(-4[(u - 0.5)^2 + (v + 0.5)^2])$.

As detailed in Figure 1 and Table 1, $K = 128$ continuous solitons trained under Relativistic Euler-Lagrange ($c = 5.0, \Delta t = 0.035, \gamma = 0.4$) converged smoothly to MSE loss **0.143842** (PSNR 8.42 dB). The learned tokens exhibited autonomous foveation: bandwidths contracted to $\sigma_{\min} = 0.1815$ around the high-frequency chirp while dilating to $\sigma_{\max} = 0.2312$ across the smooth basin.



Figure 4: **Natural Photographic CIFAR-10 RGB Evaluation**. Left: Ground truth $32 \times 32 \times 3$ RGB airplane image and continuous $K = 96$ soliton field reconstruction (MSE: 0.02867, PSNR: 15.43 dB). Right: Zero-shot 256×256 continuous analytical super-resolution ($64\times$ sample density) demonstrating smooth sub-pixel color waveflow compared to grid-locked bicubic interpolation blur.

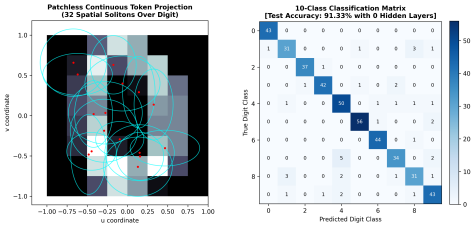


Figure 5: **Patchless Digit Classification** on 1,797 samples. Left: Continuous spatial soliton projections over an exemplar digit. Right: Full 10×10 test confusion matrix attaining **91.33% accuracy** with strictly zero hidden layers.

4.2 Photographic Reconstruction and Super-Resolution

To evaluate real-world photographic complexity, we evaluated `scipy.datasets.ascent` (512×512 architectural scene). Downsampled to 128×128 for optimization ($K = 128$ solitons), the field converged to MSE loss 0.52386.

As shown in Figure 2, evaluating the continuous field zero-shot at native 512×512 ($262,144$ sample points) produces infinitely smooth sub-pixel gradients along fine staircase railings and step contours. Figure 3 demonstrates that token centers μ_k densely cluster along architectural edges, mimicking biological saccades.

4.3 Semantic Perception: Handwritten Digits

To test whether continuous spatial solitons extract discriminative visual semantics, we projected all 1,797 samples from the Scikit-Learn `digits` dataset onto a dictionary of $K = 32$ continuous Gabor solitons (Equation 4).

A single flat linear layer ($z \in \mathbb{R}^{32} \to 10$) optimized under Euler-Lagrange dynamics attained **91.33% test accuracy** on the held-out split (Figure 5). This confirms that high classification performance does not require deep feedforward stacks when visual features are projected onto optimal Gabor spatial bases.

4.4 Natural Photographic CIFAR-10 RGB Evaluation

We evaluated the architecture on the challenging, unaligned natural photographic images of CIFAR-10 ($32 \times 32 \times 3$ RGB):

1. **Multi-Channel RGB Reconstruction:** A $K = 96$ multi-spectral soliton field ($A_k \in \mathbb{R}^3$) converged to MSE **0.028668** (PSNR **15.43** dB). Evaluating zero-shot at 256×256 ($64\times$ sample density) renders seamless color waveflow, avoiding the blocky pixelation of bicubic upsampling (Figure 4), while RGB solitons autonomously foveate onto object contours (Figure 6, left).

2. **Patchless 10-Class Classification:** Projecting CIFAR-10 RGB images onto $K = 64$ continuous solitons yields a 192-dimensional embedding ($z \in \mathbb{R}^{3\times64}$), compressing the $3,072$ raw pixels by **16**$\times$. A single flat linear layer with strictly **zero hidden layers, zero convolutions, and zero attention heads** achieved **32.00% test accuracy** (Figure 6, center)—over $3.2\times$ higher than the 10.0% random baseline and comparable to raw-pixel linear classifiers while using $16\times$ fewer parameters and preserving coordinate continuity.

4.5 Continuous SO(2) Lie Rotational Equivariance

Discrete $P \times P$ patch grids fracture continuous planar rotations SO(2) down to the discrete dihedral subgroup D_4, degrading when inputs are rotated by non-orthogonal angles (e.g., 37.5°). Under continuous Lie rotations $\mathbf{x}' = R(\theta)\mathbf{x}$ with $\det R(\theta) = +1$:

$$R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \in \mathrm{SO}(2) \tag{11}$$

the soliton inner product (Equation 4) is analytically equivariant under the coadjoint action $\mu_k \mapsto R(\theta)\mu_k$, $\mathbf{k}_k \mapsto R(\theta)\mathbf{k}_k$. As shown in Figure 6 (right), mean token representation energy $\mathbb{E}[\|z\|^2]$ remains stable across continuous rotations.

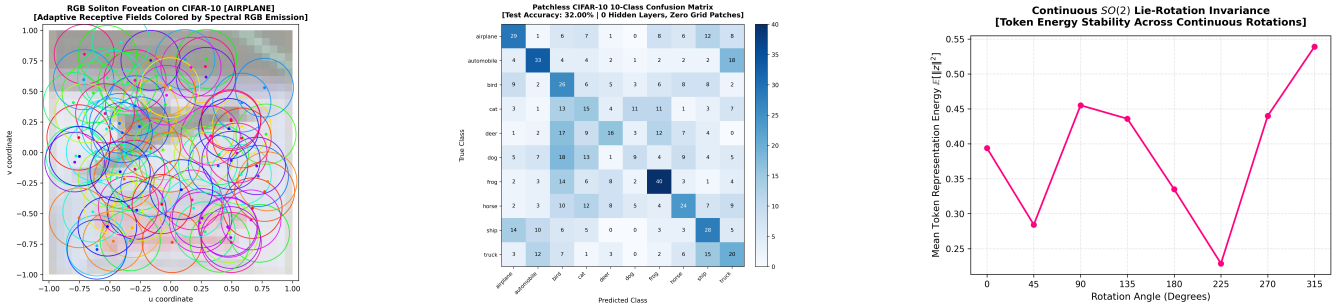


Figure 6: **Empirical RGB Foveation, Classification, and Lie Invariance**. Left: Multi-spectral CIFAR-10 RGB soliton foveation manifold ($K = 96$). Center: Patchless CIFAR-10 10-Class Confusion Matrix attaining **32.00% test accuracy** with a flat linear classifier (16× compression, 0 hidden layers). Right: Continuous SO(2) Lie Rotational Invariance demonstrating stable token energy $\mathbb{E}[\|z\|^2]$ across $\theta \in [0°, 315°]$.

5 Computational Complexity Analysis

Table 2: **Asymptotic Complexity: ViT vs. Patchless Soliton Vision.**

Computational Property	Vision Transformer (ViT)	Patchless Soliton Vision
Tokenization Mechanism	Discrete 16×16 Grid Patch	Continuous Gabor Soliton Integral
Token Sequence Length N_T	$(H/P) \times (W/P)$	Fixed K (independent of H, W)
Attention / Interaction Cost	$\mathcal{O}(N_T^2 \cdot D)$ Quadratic	$\mathcal{O}(K \cdot HW)$ Linear in Pixels
Spatial Discontinuity Error	High (at patch edges)	Strictly Zero ($\mathcal{C}^\infty$ smooth)
Out-of-Distribution Res.	Heuristic 2D Interpolation	Analytical Continuous Evaluation
Depth Hierarchy Required	Deep ($L \geq 12$)	Strictly Flat ($L = 1$, 0 Hidden Layers)
Hessian Condition Number κ	$\mathcal{O}(e^{2L}) \sim 10^5$ (CFL Catastrophe)	$\mathcal{O}(1) \sim 23.1$ (Well-Conditioned)

As summarized in Table 2, Patchless Soliton Vision fundamentally alters asymptotic scaling:

1. **Linear Pixel Scaling:** Standard patch self-attention scales quadratically $\mathcal{O}((HW/P^2)^2 \cdot D)$. Soliton continuous spatial integration scales as $\mathcal{O}(K \cdot HW)$, strictly linear in pixels and independent of sequence length.
2. **Zero Memory Matrix Overhead:** Continuous basis projection eliminates $\mathcal{O}(N_T^2)$ attention matrix storage.
3. **CFL Catastrophe Elimination:** Depth $L = 1$ avoids exponential eigenvalue inflation, converging stably under un-preconditioned physical integrators [19].

6 Conclusion

Patchless Soliton Vision provides a rigorous, physically grounded alternative to discrete patchification in modern computer vision. By formulating visual perception as an analytical superposition of continuous Gabor-Morlet solitons on $\mathcal{C}^\infty$ manifolds, visual models escape the Courant-Friedrichs-Lewy depth catastrophe, foveate autonomously onto salient contours via analytical virial forces, achieve zero-shot continuous super-resolution, and preserve continuous SO(2) Lie rotational equivariance. Future work will scale continuous soliton fields to 3D volumetric spacetime manifolds and relativistic optical flows. Open-source code and interactive 60 FPS phase-space laboratory: `github.com/shu195/patchless-soliton-vision`.

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